

RLC Network: Coupled ODE System

Study Guide — Problem 21 (Figure 9.6)

Applied Differential Equations

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1 Problem Statement

Given the RLC network governed by:

$$4I_2'(t) + 52q_1(t) = 10 \quad (1)$$

$$13I_3(t) + 52q_1(t) = 10 \quad (2)$$

$$I_1(t) = I_2(t) + I_3(t) \quad (3)$$

where $q_1(t)$ is the charge on the capacitor, $I_1(t) = q_1'(t)$, with initial conditions:

$$q_1(0) = 0, \quad I_1(0) = 0$$

Physical interpretation of the equations:

- Eq. (1): KVL around the left loop. The 4-henry inductor is on the I_2 branch, so the inductive drop is $4I_2'$. Same capacitor q_1/C appears in both loops.
- Eq. (2): KVL around the right loop. The 13-ohm resistor carries I_3 .
- Eq. (3): KCL at the top node.

Goal: Solve for $I_1(t)$, $I_2(t)$, $I_3(t)$.

Hint (from text): Differentiate (1) and (2), eliminate I_1 using (3), form a normal system with $x_1 = I_2$, $x_2 = I_2'$, $x_3 = I_3$.

2 Prerequisite Theorems and Definitions

Definition 2.1: RLC Circuit Element Relations

- **Capacitor:** $V_C = q/C$, where $q(t) = \int I_1 dt$ and $I_1 = q'(t)$.
- **Inductor on branch k :** $V_L = L I_k'(t)$. The derivative is of the branch current, not the source current.
- **Resistor:** $V_R = R I(t)$.

Definition 2.2: Kirchhoff's Laws

- **KVL:** $\sum V_k = 0$ around any closed loop.
- **KCL:** $\sum I_k = 0$ at any node. Here: $I_1 = I_2 + I_3$.

Definition 2.3: Normal Form

A system is in *normal form* if:

$$\mathbf{x}'(t) = A \mathbf{x}(t) + \mathbf{f}(t), \quad A \text{ constant matrix.}$$

This form is required for the eigenvalue solution method.

Theorem 2.1: Differentiation to Eliminate Charge

If an equation contains $q_1(t)$, differentiate with respect to t . Since $q_1'(t) = I_1(t)$, every q_1 term becomes I_1 , removing the integral from the system.

Theorem 2.2: Eigenvalue Method for $\mathbf{x}' = A\mathbf{x}$

1. Compute $\det(A - \lambda I) = 0$ to find eigenvalues λ_i .
2. For each λ_i , solve $(A - \lambda_i I)\mathbf{v}_i = 0$ for eigenvector $\mathbf{v}_i$.
3. Real λ : contributes $c_i e^{\lambda t} \mathbf{v}_i$.
4. Complex pair $\alpha \pm \beta i$ with eigenvector $\mathbf{a} + i\mathbf{b}$: contributes

$$e^{\alpha t} [c_2(\mathbf{a} \cos \beta t - \mathbf{b} \sin \beta t) + c_3(\mathbf{b} \cos \beta t + \mathbf{a} \sin \beta t)].$$

5. Apply initial conditions to determine constants c_i .

3 Step-by-Step Solution

3.1 Step 1: Differentiate Both Loop Equations

Differentiate (1) (using $q_1' = I_1$):

$$4I_2''(t) + 52I_1(t) = 0$$

Differentiate (2):

$$13I_3'(t) + 52I_1(t) = 0$$

Why? The original equations mix $q_1(t)$ (an integral quantity) with currents. Differentiating converts $q_1' \rightarrow I_1$, making every term a derivative of a current, enabling normal form.

3.2 Step 2: Substitute $I_1 = I_2 + I_3$ (KCL)

Replace I_1 in both differentiated equations:

$$4I_2'' + 52(I_2 + I_3) = 0 \implies I_2'' + 13I_2 + 13I_3 = 0$$

$$13I_3' + 52(I_2 + I_3) = 0 \implies I_3' = -4I_2 - 4I_3$$

Why eliminate I_1 ? After differentiation, I_1 appears in both equations but is fully determined by I_2 and I_3 via KCL. Substituting reduces the system to two independent unknowns.

3.3 Step 3: Write as First-Order Normal System

Let $x_1 = I_2$, $x_2 = I_2'$, $x_3 = I_3$. Then:

$$\begin{aligned}x_1' &= x_2 \\x_2' &= -13x_1 - 13x_3 \\x_3' &= -4x_1 - 4x_3\end{aligned}$$

In matrix form $\mathbf{x}' = A\mathbf{x}$:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ -13 & 0 & -13 \\ -4 & 0 & -4 \end{pmatrix}$$

This is a homogeneous system. The 10-volt constant source differentiated to zero. Its physical effect is encoded entirely in the initial conditions.

3.4 Step 4: Determine Initial Conditions for the Normal System

Given: $q_1(0) = 0$, $I_1(0) = 0$.

Find $I_3(0)$: Evaluate (2) at $t = 0$:

$$13I_3(0) + 52 \cdot 0 = 10 \implies I_3(0) = \frac{10}{13}$$

Find $I_2(0)$: From KCL $I_1(0) = I_2(0) + I_3(0) = 0$:

$$I_2(0) = -\frac{10}{13}$$

Find $I_2'(0)$: Evaluate (1) at $t = 0$:

$$4I_2'(0) + 52 \cdot 0 = 10 \implies I_2'(0) = \frac{5}{2}$$

$$\mathbf{x}(0) = \begin{pmatrix} \frac{10}{13} \\ -\frac{10}{13} \\ \frac{5}{2} \end{pmatrix}$$

Key point: $I_2'(0)$ is extracted from the *original* equation (1) before differentiation. After differentiation, the constant source vanishes and this information is lost.

3.5 Step 5: Find Eigenvalues of A

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 & 0 \\ -13 & -\lambda & -13 \\ -4 & 0 & -4 - \lambda \end{pmatrix} = 0$$

Expand along the first row:

$$-\lambda[(-\lambda)(-4 - \lambda)] - 1[(-13)(-4 - \lambda)] = 0$$

$$-\lambda(\lambda^2 + 4\lambda) - (52 + 13\lambda) = 0$$

$$-\lambda^3 - 4\lambda^2 - 13\lambda = 0$$

$$-\lambda(\lambda^2 + 4\lambda + 13) = 0$$

Solving $\lambda^2 + 4\lambda + 13 = 0$:

$$\lambda = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm 3i$$

$$\lambda_1 = 0, \quad \lambda_{2,3} = -2 \pm 3i$$

Physical interpretation:

- $\lambda_1 = 0$: DC steady-state constant term.
- $\lambda_{2,3} = -2 \pm 3i$: Damped oscillation. The -2 (negative real part) comes from resistive dissipation (13 ohms). The $3i$ gives oscillation at 3 rad/s from the LC reactive pair. Contrast: a purely reactive circuit (no resistor) would give $\pm i\beta$ with no decay.

3.6 Step 6: Find Eigenvectors

For $\lambda_1 = 0$: Solve $A\mathbf{v} = 0$:

Row 1: $v_2 = 0$. Row 3: $v_3 = -v_1$. Let $v_1 = 1$:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

For $\lambda_2 = -2 + 3i$: Solve $(A - \lambda_2 I)\mathbf{v} = 0$:

$$\begin{pmatrix} 2-3i & 1 & 0 \\ -13 & 2-3i & -13 \\ -4 & 0 & -2-3i \end{pmatrix} \mathbf{v} = 0$$

Row 1: $v_2 = (-2 + 3i)v_1$.

Row 3: $v_3 = \frac{-4}{2+3i}v_1 = \frac{-4(2-3i)}{13}v_1 = \frac{-8+12i}{13}v_1$.

Let $v_1 = 13$:

$$\mathbf{v}_2 = \begin{pmatrix} 13 \\ -26 + 39i \\ -8 + 12i \end{pmatrix} = \underbrace{\begin{pmatrix} 13 \\ -26 \\ -8 \end{pmatrix}}_{\mathbf{a}} + i \underbrace{\begin{pmatrix} 0 \\ 39 \\ 12 \end{pmatrix}}_{\mathbf{b}}$$

3.7 Step 7: Write the General Solution

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 + e^{-2t} [c_2 (\mathbf{a} \cos 3t - \mathbf{b} \sin 3t) + c_3 (\mathbf{b} \cos 3t + \mathbf{a} \sin 3t)]$$

Expanding:

$$I_2(t) = c_1 + 13e^{-2t}(c_2 \cos 3t + c_3 \sin 3t)$$

$$I_2'(t) = e^{-2t} [(-26c_2 + 39c_3) \cos 3t + (-39c_2 - 26c_3) \sin 3t]$$

$$I_3(t) = -c_1 + e^{-2t} [(-8c_2 + 12c_3) \cos 3t + (-12c_2 - 8c_3) \sin 3t]$$

3.8 Step 8: Apply Initial Conditions

Set $t = 0$:

$$\text{(IC1):} \quad c_1 + 13c_2 = -\frac{10}{13} \quad (4)$$

$$\text{(IC2):} \quad -26c_2 + 39c_3 = \frac{5}{2} \quad (5)$$

$$\text{(IC3):} \quad -c_1 - 8c_2 + 12c_3 = \frac{10}{13} \quad (6)$$

Add (IC1) and (IC3):

$$5c_2 + 12c_3 = 0 \implies c_2 = -\frac{12}{5}c_3$$

Substitute into (IC2):

$$-26\left(-\frac{12}{5}c_3\right) + 39c_3 = \frac{5}{2} \implies \frac{507}{5}c_3 = \frac{5}{2} \implies c_3 = \frac{25}{1014}$$

Back-substitute:

$$\begin{aligned} c_2 &= -\frac{12}{5} \cdot \frac{25}{1014} = -\frac{10}{169} \\ c_1 &= -\frac{10}{13} - 13\left(-\frac{10}{169}\right) = -\frac{10}{13} + \frac{10}{13} = 0 \end{aligned}$$

$$\boxed{c_1 = 0, \quad c_2 = -\frac{10}{169}, \quad c_3 = \frac{25}{1014}}$$

3.9 Step 9: Final Currents

Compute $I_2(t)$ (with $c_1 = 0$):

$$I_2(t) = 13e^{-2t} \left(-\frac{10}{169} \cos 3t + \frac{25}{1014} \sin 3t \right)$$

$$\boxed{I_2(t) = e^{-2t} \left(-\frac{10}{13} \cos 3t + \frac{25}{78} \sin 3t \right)}$$

Compute $I_3(t)$ coefficients:

$$\begin{aligned} -8c_2 + 12c_3 &= \frac{80}{169} + \frac{300}{1014} = \frac{480+300}{1014} = \frac{780}{1014} = \frac{10}{13} \\ -12c_2 - 8c_3 &= \frac{120}{169} - \frac{200}{1014} = \frac{720-200}{1014} = \frac{520}{1014} = \frac{20}{39} \end{aligned}$$

$$\boxed{I_3(t) = e^{-2t} \left(\frac{10}{13} \cos 3t + \frac{20}{39} \sin 3t \right)}$$

Compute $I_1(t) = I_2(t) + I_3(t)$:

$$I_1(t) = e^{-2t} \left[\left(-\frac{10}{13} + \frac{10}{13} \right) \cos 3t + \left(\frac{25}{78} + \frac{40}{78} \right) \sin 3t \right]$$

$$\boxed{I_1(t) = \frac{5}{6} e^{-2t} \sin 3t}$$

Verification: $I_1(0) = \frac{5}{6} \cdot 0 = 0 \checkmark$

4 General Strategy (Transferable Algorithm)

Key Idea 4.1: Algorithm for Any Coupled RLC System

- Step 1. Label branches.** Assign a current variable to each distinct branch. The inductor voltage is $L I'_k$ where k is the inductor's branch, not the node/source current.
- Step 2. Write KVL** for each independent loop using element relations.
- Step 3. Write KCL** at each node.
- Step 4. Differentiate** equations containing $q(t)$, using $q' = I_{\text{node}}$.
- Step 5. Substitute KCL** to eliminate one current and reduce unknowns.
- Step 6. Introduce state variables** and write $\mathbf{x}' = A\mathbf{x} + \mathbf{b}$.
- Step 7. Recover ICs** from original (pre-differentiated) equations at $t = 0$.
- Step 8. Find eigenvalues** of A ; classify solution type.
- Step 9. Find eigenvectors**; split complex ones into real and imaginary parts $\mathbf{a}$, $\mathbf{b}$.
- Step 10. Apply ICs** to solve for c_i .
- Step 11. Recover eliminated currents** via KCL.

4.1 Logical Dependency Map

$$\underbrace{\text{KVL} + \text{element laws}}_{\text{eqns with } q_1} \xrightarrow{\frac{d}{dt}} \underbrace{\text{eqns in } I, I', I''}_{\text{no integrals}} \xrightarrow{\text{KCL sub}} \underbrace{\mathbf{x}' = A\mathbf{x}}_{\text{normal form}} \xrightarrow{\text{eig}(A)} \underbrace{I_2(t), I_3(t)}_{\text{state soln}} \xrightarrow{\text{KCL}} I_1(t)$$

5 Active Recall Questions

Active Recall 5.1: Q1 — Why does the first loop equation have I'_2 , not I'_1 ?

Because KVL traces voltage drops around the left loop. The 4-henry inductor is on the I_2 branch (the inductor branch), so its voltage drop is $4I'_2$. The current I_1 is the total node/source current, not the inductor branch current. Confusing branch current with node current is the canonical error in multi-branch RLC problems.

Active Recall 5.2: Q2 — What do eigenvalues $-2 \pm 3i$ imply physically?

The factor e^{-2t} is exponential decay from resistive dissipation (13 ohms). The factor $\cos 3t$, $\sin 3t$ is oscillation at 3 rad/s from the LC reactive elements. The combination is damped oscillation: all transient currents $\rightarrow 0$ as $t \rightarrow \infty$. A purely imaginary pair ($\pm 3i$, no real part) would indicate an undamped (lossless) circuit.

Active Recall 5.3: Q3 — Why is $c_1 = 0$?

c_1 multiplies the $\lambda_1 = 0$ eigenvector, which represents a DC steady-state offset. With $q_1(0) = 0$ and $I_1(0) = 0$, the initial conditions force $c_1 = 0$. Physically, the capacitor blocks DC in steady state, so no constant current component survives.

Active Recall 5.4: Q4 — How is $I'_2(0)$ obtained?

Evaluate (1) at $t = 0$ before differentiating: $4I'_2(0) + 52q_1(0) = 10$. With $q_1(0) = 0$, get $I'_2(0) = 5/2$. After differentiation, the source term vanishes, so this value is not recoverable from the differentiated system.

Active Recall 5.5: Q5 — Why does differentiation produce a homogeneous system here?

The source (10 V) is a constant. $\frac{d}{dt}[10] = 0$. Any constant right-hand side differentiates to zero, yielding a homogeneous system. The source's physical contribution is entirely transferred into the initial conditions (it determines $I'_2(0)$ and $I_3(0)$).

Active Recall 5.6: Q6 — What changes if the source is $f(t)$ (time-varying)?

Differentiating gives $f'(t) \neq 0$ on the right side, yielding $\mathbf{x}' = A\mathbf{x} + \mathbf{g}(t)$ (non-homogeneous). Find a particular solution $\mathbf{x}_p(t)$ via undetermined coefficients (if f is polynomial, exponential, or sinusoidal) or variation of parameters (general), then add to the homogeneous solution.

6 Summary Table

Concept	Formula	Role here
Inductor voltage (branch k)	$V_L = L I'_k$	k = inductor branch, not node
Capacitor charge relation	$q'(t) = I_1(t)$	Differentiates q_1 out of equations
KCL	$I_1 = I_2 + I_3$	Eliminates I_1 ; recovers it at end
Normal form	$\mathbf{x}' = A\mathbf{x}$	Required for eigenvalue method
Characteristic polynomial	$\det(A - \lambda I) = 0$	Finds eigenvalues
$\lambda = 0$	constant	DC steady-state component
$\lambda = \alpha \pm \beta i$	$e^{\alpha t}(\cos \beta t, \sin \beta t)$	Damped oscillation
$\lambda = \pm \beta i$	$\cos \beta t, \sin \beta t$	Undamped oscillation
Complex eigenvector split	$\mathbf{v} = \mathbf{a} + i\mathbf{b}$	Produces real solution basis

7 Generalizations and Pitfalls

7.1 Critical Distinction: Branch Current vs. Node Current

The canonical error: writing LI'_1 for the inductor voltage when the inductor is on branch 2. The inductor voltage is $L I'_k$ for the branch carrying the inductor. Always label circuit branches and assign a current variable to each before writing KVL.

7.2 Recovering $I'_k(0)$ Before Differentiating

Before differentiating, evaluate the loop equations at $t = 0$ with the given physical initial values. After differentiation, the constant source vanishes and this information is unrecoverable from the system alone.

7.3 When $c_1 \neq 0$

If the DC eigenvector component is nonzero, the circuit has a nonzero steady-state current. This occurs when the source drives a purely resistive path with no capacitor in series. Check whether every source-to-ground path passes through the capacitor; if yes, $c_1 = 0$ at steady state.

7.4 Repeated Eigenvalues

If λ has algebraic multiplicity $>$ geometric multiplicity, include a generalized eigenvector $\mathbf{w}$ satisfying $(A - \lambda I)\mathbf{w} = \mathbf{v}$:

$$\mathbf{x} \supset e^{\lambda t}(c_1 \mathbf{v} + c_2(t\mathbf{v} + \mathbf{w}))$$

7.5 Non-Homogeneous After Differentiation

If $f(t)$ is time-varying, the system $\mathbf{x}' = A\mathbf{x} + \mathbf{g}(t)$ requires a particular solution. Use undetermined coefficients when $\mathbf{g}(t)$ is a polynomial, exponential, or sinusoidal; use variation of parameters in the general case. Add $\mathbf{x}_p$ to the homogeneous solution before applying initial conditions.